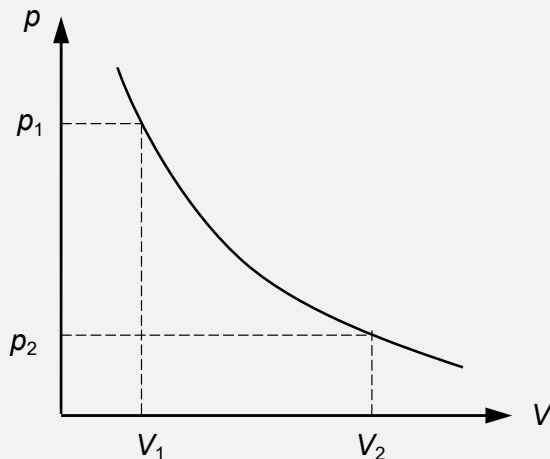


15. An ideal gas, contained in a cylinder by a frictionless piston, is allowed to expand from volume V_1 , at a pressure p_1 , to volume V_2 , at pressure p_2 . Its temperature is kept constant throughout. The work done by the gas is
- A** Zero, because it obeys Boyle's Law and therefore $p_2V_2 - p_1V_1 = 0$.
- B** Negative, because the pressure has decreased and so the force on the piston has been diminishing.
- C** Zero, because it has been kept at constant temperature and so its internal energy is unchanged.
- D** Positive, because the volume has increased.

Answer: D



work done = area under p - V graph from V_1 to V_2

- A:** Incorrect. Although gas obey Boyle's law, area under graph $\neq 0$
- B:** Incorrect. Gas expand indicate work done by gas is positive.
- C:** Incorrect. Although temperature is constant during expansion, area under graph $\neq 0$

Note: When T is constant, $\Delta U = 0$ for ideal gas, so $q = -w$ $\square$ q = work done by gas